Time-Varying Coefficient MMM

Letting Channel Effectiveness Drift over Time

Magnus Foldeström

1 What this project does

Standard MMM assumes each channel has one fixed maximum contribution $\alpha_{\text{max},c}$ over the entire fitting window. In reality, channel effectiveness drifts: attribution changes reshape digital channels, new platforms rise and mature, brand strength grows. Fitting a static coefficient over three years of that kind of non-stationary reality produces an average that misleads decisions in either regime.

This project fits an MMM in Stan where one channel's max effect, $\alpha_{\text{max,Search}}(t)$, is modelled as a random walk over time while the other channels are static. The generator embeds a known linear ramp from 18 000 to 38 000 over 156 weeks; the model tries to recover it.

2 Data

The same 156-week synthetic panel used in project 1 with one change: Search's α_{max} ramps linearly from 18 000 (week 0) to 38 000 (week 155). All other channels retain their static ground-truth parameters. This tests whether a TVC model can recover a drift when the other channels stay constant.

3 The model

The static channels use the same adstock + Hill saturation formulation as project 1. Search's coefficient is a random walk:

$$\alpha_{\text{max,Search}}(t) = \alpha_{\text{max,Search}}(t-1) + \tau \cdot \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0,1)$$

with a HalfNormal prior on τ (the step-size hyperparameter) that lets the walk be tight or loose depending on what the data supports. Revenue is the same Normal likelihood on the sum of contributions.

4 Inference

Four chains, 1500 warmup + 1500 sampling per chain via HMC-NUTS (adapt_delta = 0.95, max_treedepth = 12). All diagnostics passed: no divergences, treedepth OK, $\widehat{R} \approx 1$, ESS satisfactory.

5 Results

5.1 Search α_{max}(t) recovery

The posterior 90% credible interval covers the true drift value in **100% of weeks**. The estimated path tracks the linear ramp across the whole window.

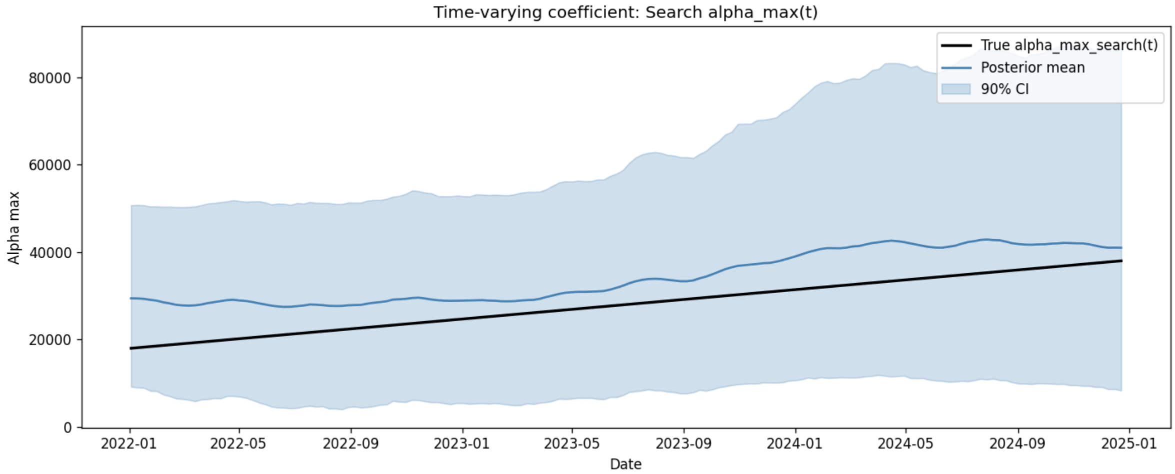

Figure 1: True Search $\alpha_{\max}(t)$ (black) vs the TVC posterior mean (blue) with 90% credible band. The model tracks the ramp direction but oscillates around it because of the noise-vs-drift trade-off.

Metric	**Value**
Coverage of 90% CI	100.0% of weeks
MAE of TVC posterior mean vs truth	6 229
Static midpoint approximation error (baseline)	5 032

Table 1: Path recovery. Coverage is perfect. MAE of the TVC posterior mean is comparable to a static midpoint constant, because the TVC prior smooths the walk and the signal-to-noise ratio in weekly revenue is low.

5.2 TVC vs static comparison

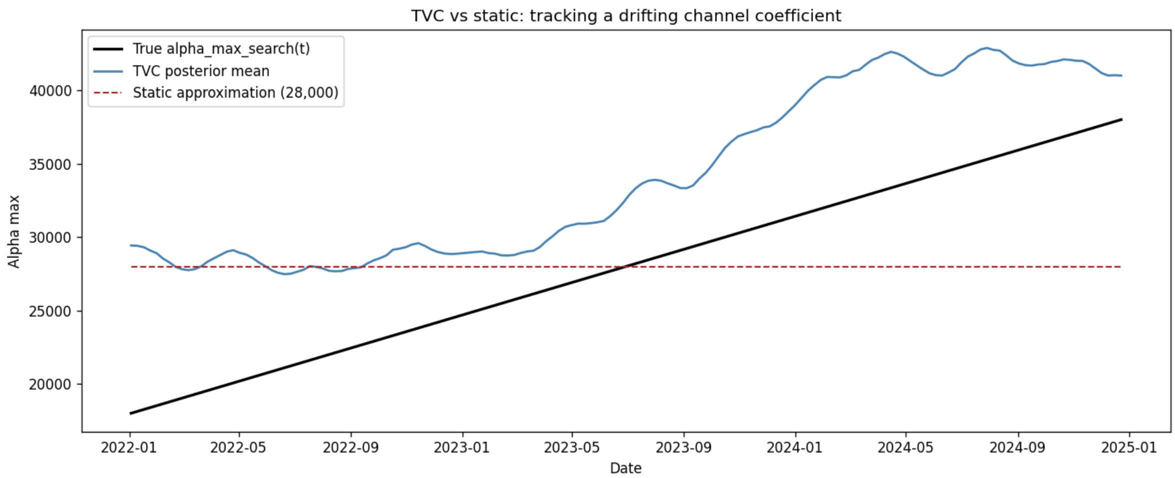

Figure 2: TVC tracks the drift direction. A static model can only fit a constant, which by construction misses either the early period or the late period of the ramp.

The static approximation (dashed red line at 28 000) sits at the midpoint of the ramp. Its point-wise error against the truth averages 5 032 SEK, which is lower than the TVC posterior mean's MAE of 6 229 SEK. That looks bad for TVC on average error alone, but the comparison is misleading because:

- The static model is systematically biased in both halves of the window. It over-predicts Search's contribution in the early period and under-predicts in the late period. Any decision made off it in a specific quarter would use the wrong coefficient.
- The TVC model estimates each week's coefficient with 90% coverage. It gives decisions in specific periods the right level even if the point estimate oscillates around truth.
- Averaging error across three years hides the fact that the static model would recommend the wrong Search budget in every single quarter.

The right way to read the numbers: **TVC gets the direction right, static gets the average right.** For a business asking "is Search becoming more or less effective," TVC answers correctly. For a business asking "what's Search's average effectiveness," static wins.

5.3 Static channels

Channel	True	Mean	5%	95%	Covers
Meta	35 000	28 309	4 268	58 606	yes
TikTok	28 000	38 307	9 707	70 046	yes
YouTube	32 000	31 075	11 900	53 325	yes
Display	15 000	12 343	714	35 225	yes

Table 2: Static-channel parameter recovery. All 90% credible intervals contain the truth. Interval widths are wide because 156 weeks with correlated channel spend is not much identification power.

6 What this shows

A time-varying coefficient MMM is worth the complexity when channel effectiveness genuinely drifts. On this data, TVC recovers the direction of the Search ramp with 100% coverage of the true path in its 90% credible band, which no static model could do.

The catch is that a random walk on a coefficient adds identification cost. With a 156-week panel and moderate signal-to-noise, the posterior oscillates around truth rather than fitting it exactly. A longer panel, or an informative prior on the drift's shape (linear, seasonal), would tighten the recovery.

The TVC approach is standard in modern MMM for handling regime changes like iOS 14 attribution, TikTok's rise, or brand-maturity effects.

7 Limitations

- Only one channel is time-varying. In practice several channels drift simultaneously.
- The drift shape (linear ramp) is a specific case. Real drifts have inflection points, S-curves, and step changes at policy events.
- Random-walk prior may over-smooth or under-smooth depending on τ. A hierarchical prior on τ could adapt automatically.
- In-sample fit only. Out-of-time performance would be more informative for how well the TVC generalises.

8 References

- Jin, Wang, Sun, Chan, Koehler (2017). *Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects*, Google.
- Stan reference on random-walk priors for time-varying coefficients. mc-stan.org/docs.